

LIU HENGZHI

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EDUCATION

National University of Singapore <i>Bachelor of Computing in Computer Science (Tuition Grant)</i>	2026 – June 2029
Temasek Polytechnic, Singapore <i>Diploma in Biomedical Engineering with Merit</i> Awarded Polytechnic Scholarship, Course Bronze Medalist, and Director's List honors.	2023 – 2026

SKILLS

Programming Languages: Python, C++, C, SQL

Backend & Systems: FastAPI, System Design Basics, LLM Agents, AWS (Aurora), Embedded Systems, FreeRTOS

Tools & Utilities: Git/GitHub, Linux, RESTful APIs, MySQL, CAD, Adobe Suite

TECHNICAL PROJECTS

DTx Cognitive SaaS (ADHD Therapeutics) <i>Full-Stack Developer</i>	2026 <i>React, FastAPI, AWS Aurora, LLM Agents</i> - Secured clinical ADHD SaaS platform endpoints by implementing AWS token-based authentication. - Optimized database query performance by batching telemetry and tuning AWS Aurora overhead. - Accelerated diagnostic report generation by building a FastAPI backend with local caching.
Smart Mahjong Play-Board <i>Independent Systems Developer</i>	2026 <i>C++, ESP32, PCB Design, Embedded Firmware</i> - Engineered an ESP32-based smart play-board featuring custom PCB circuit integration. - Established low-latency hardware-software communication by designing customized serial protocols in embedded firmware.

WORK EXPERIENCE

National University of Singapore (iHealthtech) <i>Research Assistant Intern</i>	Mar 2025 – Jul 2025 <i>Singapore</i> - Generated clean research datasets by executing microscale biological testing procedures. - Ensured source data accuracy by executing routine validation tests on laboratory instruments. - Enhanced data auditability by setting up structured digital documentation guidelines.
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CO-CURRICULAR ACTIVITIES

WorldSkills Singapore (Additive Manufacturing)	2024 Won 1st Place (TP Selection Round) by modeling and optimizing 3D printable products.
Electronic Design & Robotics Competitions	2024 Developed embedded logic systems and real-time control algorithms for robotics challenges.